

Database System Engineering and Distributed Backend Development

WEEK-4: BookFlow Intelligent Catalog & API Blueprint

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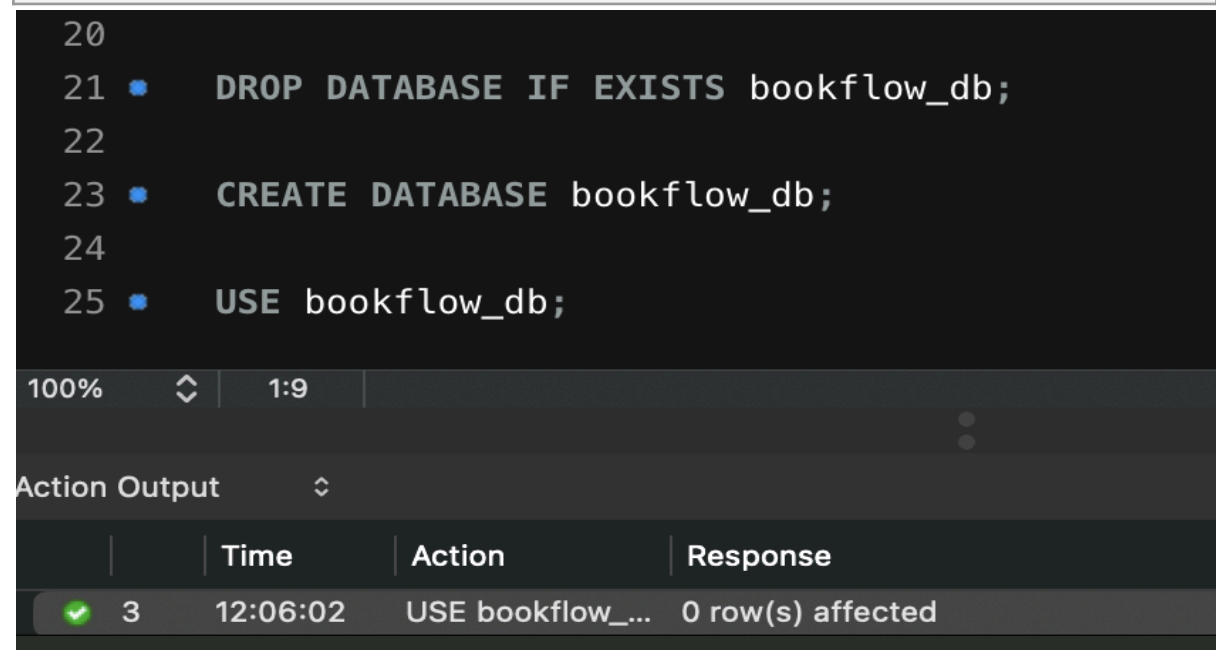
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Step 1 — Create Database

```
DROP DATABASE IF EXISTS bookflow_db;  
CREATE DATABASE bookflow_db;  
USE bookflow_db;
```



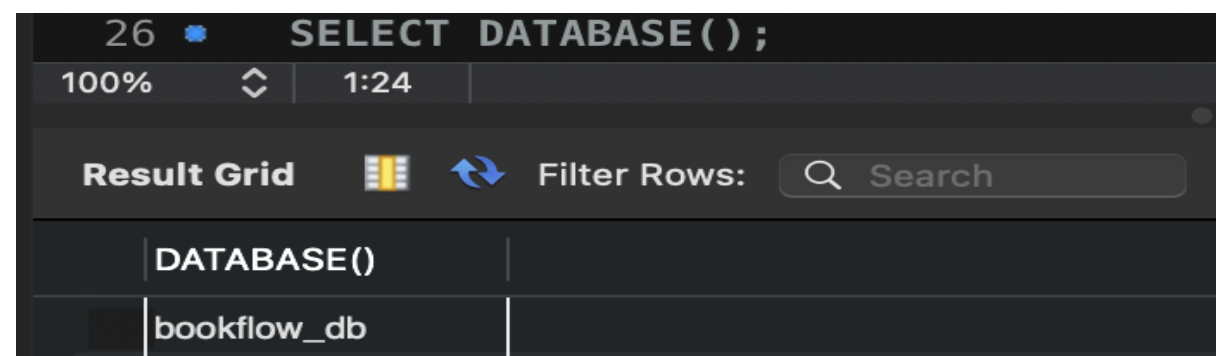
The screenshot shows a SQL IDE interface. The top panel displays the following SQL commands:

```
20  
21 DROP DATABASE IF EXISTS bookflow_db;  
22  
23 CREATE DATABASE bookflow_db;  
24  
25 USE bookflow_db;
```

Below the code editor, the 'Action Output' panel is visible. It shows a table with the following data:

	Time	Action	Response
✓ 3	12:06:02	USE bookflow_...	0 row(s) affected

```
SELECT DATABASE();
```



The screenshot shows a SQL IDE interface. The top panel displays the following SQL command:

```
26 SELECT DATABASE();
```

Below the code editor, the 'Result Grid' panel is visible. It shows a table with the following data:

DATABASE()
bookflow_db

Step 2 — Create Books Table

```
CREATE TABLE Books (  
    book_id INT AUTO_INCREMENT PRIMARY KEY,  
    title VARCHAR(200) NOT NULL,  
    author VARCHAR(150) NOT NULL,  
    genre VARCHAR(100),  
    description TEXT,  
    published_year INT  
);
```

21 CREATE TABLE Books (
22 book_id INT AUTO_INCREMENT PRIMARY KEY,
23 title VARCHAR(200) NOT NULL,
24 author VARCHAR(150) NOT NULL,
25 genre VARCHAR(100),
26 description TEXT,
27 published_year INT
28);
29
30

0% 1:12

ion Output

	Time	Action	Response
✓ 5	12:20:11	CREATE TABLE...	0 row(s) affected

Check the structure:

```
DESCRIBE Books;
```

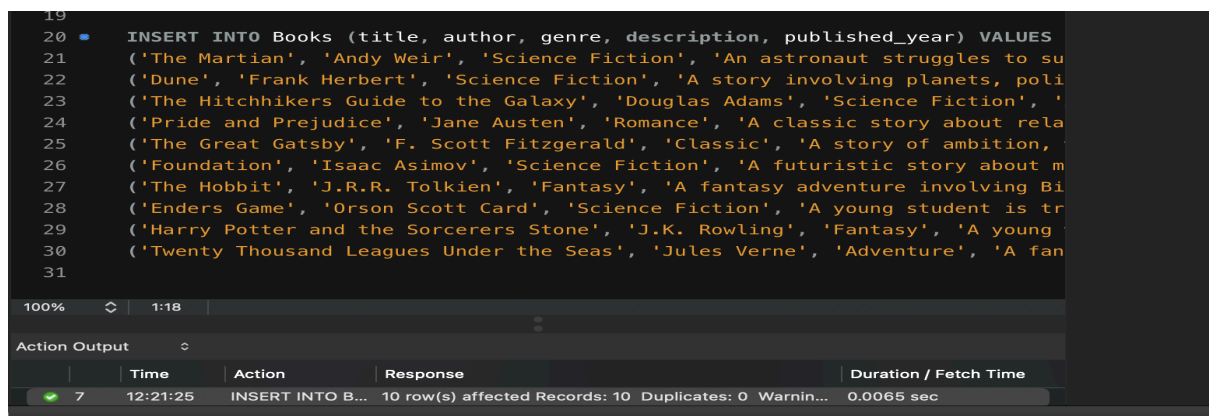
19
20 DESCRIBE Books;
100% 1:16

Result Grid Filter Rows: Search Export:

	Field	Type	Null	Key	Default	Extra	
	book_id	int	NO	PRI	NULL	auto_increment	
	title	varchar(200)	NO		NULL		
	author	varchar(150)	NO		NULL		
	genre	varchar(100)	YES		NULL		
	description	text	YES		NULL		
	published_year	int	YES		NULL		

Step 3 — Insert 10 Books

```
INSERT INTO Books (title, author, genre, description, published_year) VALUES
('The Martian', 'Andy Weir', 'Science Fiction', 'An astronaut struggles to survive
alone on Mars.', 2011),
('Dune', 'Frank Herbert', 'Science Fiction', 'A story involving planets, politics,
survival and space.', 1965),
('The Hitchhikers Guide to the Galaxy', 'Douglas Adams', 'Science Fiction', 'A
humorous adventure across space and alien worlds.', 1979),
('Pride and Prejudice', 'Jane Austen', 'Romance', 'A classic story about
relationships and social expectations.', 1813),
('The Great Gatsby', 'F. Scott Fitzgerald', 'Classic', 'A story of ambition,
wealth, love and the American dream.', 1925),
('Foundation', 'Isaac Asimov', 'Science Fiction', 'A futuristic story about
mathematics, civilization and space.', 1951),
('The Hobbit', 'J.R.R. Tolkien', 'Fantasy', 'A fantasy adventure involving Bilbo
Baggins and a dangerous journey.', 1937),
('Enders Game', 'Orson Scott Card', 'Science Fiction', 'A young student is trained
for an interstellar conflict.', 1985),
('Harry Potter and the Sorcerers Stone', 'J.K. Rowling', 'Fantasy', 'A young wizard
begins his magical education.', 1997),
('Twenty Thousand Leagues Under the Seas', 'Jules Verne', 'Adventure', 'A fantastic
underwater adventure aboard a mysterious submarine.', 1870);
```



```
19
20 INSERT INTO Books (title, author, genre, description, published_year) VALUES
21 ('The Martian', 'Andy Weir', 'Science Fiction', 'An astronaut struggles to su
22 ('Dune', 'Frank Herbert', 'Science Fiction', 'A story involving planets, poli
23 ('The Hitchhikers Guide to the Galaxy', 'Douglas Adams', 'Science Fiction', '
24 ('Pride and Prejudice', 'Jane Austen', 'Romance', 'A classic story about rela
25 ('The Great Gatsby', 'F. Scott Fitzgerald', 'Classic', 'A story of ambition,
26 ('Foundation', 'Isaac Asimov', 'Science Fiction', 'A futuristic story about m
27 ('The Hobbit', 'J.R.R. Tolkien', 'Fantasy', 'A fantasy adventure involving Bi
28 ('Enders Game', 'Orson Scott Card', 'Science Fiction', 'A young student is tr
29 ('Harry Potter and the Sorcerers Stone', 'J.K. Rowling', 'Fantasy', 'A young
30 ('Twenty Thousand Leagues Under the Seas', 'Jules Verne', 'Adventure', 'A fan
31
```

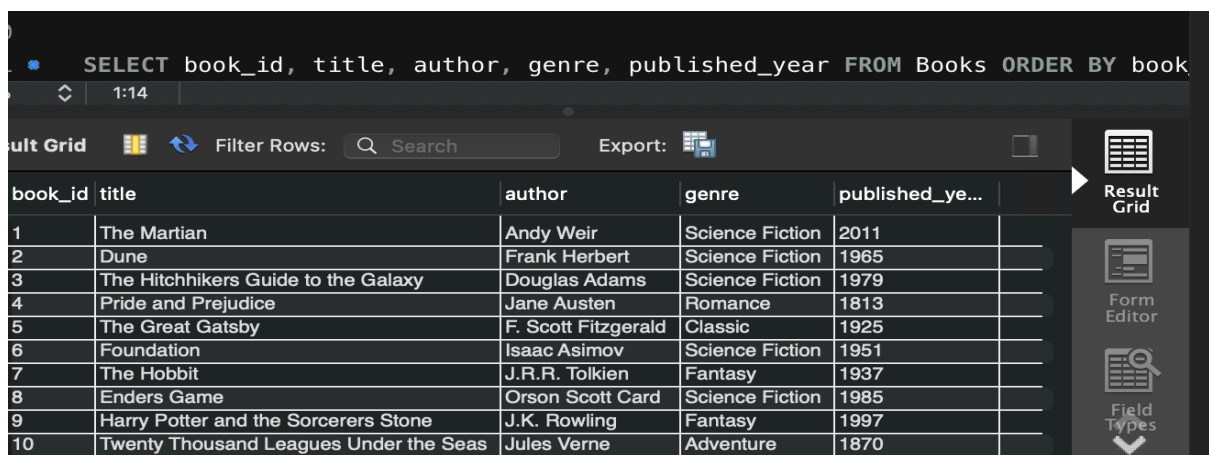
100% 1:18

Action Output

	Time	Action	Response	Duration / Fetch Time
7	12:21:25	INSERT INTO B...	10 row(s) affected Records: 10 Duplicates: 0 Warnin...	0.0065 sec

Step 4 — Display All Books

```
SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id;
```



```
SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book
```

1:14

ult Grid Filter Rows: Search Export:

book_id	title	author	genre	published_ye...
1	The Martian	Andy Weir	Science Fiction	2011
2	Dune	Frank Herbert	Science Fiction	1965
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	Pride and Prejudice	Jane Austen	Romance	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6	Foundation	Isaac Asimov	Science Fiction	1951
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	Enders Game	Orson Scott Card	Science Fiction	1985
9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997
10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870

Result Grid Form Editor Field Types

Step 5 — Add the Embedding Column (Semantic Search Setup)

ALTER TABLE Books
ADD COLUMN embedding JSON;

23

24 • ALTER TABLE Books

25 ADD COLUMN embedding JSON;

100%

1:18

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 9	12:32:06	ALTER TABLE...	0 row(s) affected Records: 0 Duplicates: 0 Warnings...	0.0095 sec

DESCRIBE Books;

24

25 • DESCRIBE Books;

100%

1:22

Result Grid

Filter Rows:

Search

Export:

	Field	Type	Null	Key	Default	Extra	
	book_id	int	NO	PRI	NULL	auto_increment	
	title	varchar(200)	NO		NULL		
	author	varchar(150)	NO		NULL		
	genre	varchar(100)	YES		NULL		
	description	text	YES		NULL		
	published_year	int	YES		NULL		
	embedding	json	YES		NULL		

Step 6 — Insert Mock Vectors for Each Book

UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;

```
24
25 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
26 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
27 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
28 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
29 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
30 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
31 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
32 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
33 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
34 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
35
100% 1:20
Action Output
Time Action Response Duration / Fetch Time
20 12:33:52 UPDATE Books... 1 row(s) affected Rows matched: 1 Changed: 1 Warni... 0.00053 sec
```

Step 7 — Verify All Vectors

```
SELECT book_id, title, embedding FROM Books ORDER BY book_id;
```

```
22
23 • SELECT book_id, title, embedding FROM Books ORDER BY book_id;
```

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]
9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]

Step 8 — Similarity Search: L2 Distance ("closest" books)

Search vector used = $[0.90, 0.80, 0.20]$ → represents "Stories about space". Smaller distance = more similar.

```
SELECT
  book_id, title, author, genre,
  ROUND(
    SQRT(
      POW(CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) - 0.90, 2) +
      POW(CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) - 0.80, 2) +
      POW(CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) - 0.20, 2)
    ), 4
  ) AS l2_distance
FROM Books
WHERE embedding IS NOT NULL
ORDER BY l2_distance ASC
LIMIT 3;
```

Conclusion: The 3 closest books are The Martian, Enders Game, and Dune.

```

21 SELECT
22     book_id, title, author, genre,
23     ROUND(
24         Sqrt(
25             POW(CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) - 0.90,
26             POW(CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) - 0.80,
27             POW(CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) - 0.20,
28             ), 4
29         ) AS l2_distance
30 FROM Books
31 WHERE embedding IS NOT NULL
32 ORDER BY l2_distance ASC
33 LIMIT 3;
34

```

100% 1:34

Result Grid Filter Rows: Search Export:

	book_id	title	embedding
	1	The Martian	[0.90, 0.80, 0.20]
	2	Dune	[0.85, 0.75, 0.25]
	3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
	4	Pride and Prejudice	[0.10, 0.20, 0.90]
	5	The Great Gatsby	[0.15, 0.30, 0.80]
	6	Foundation	[0.95, 0.65, 0.20]
	7	The Hobbit	[0.30, 0.75, 0.90]
	8	Enders Game	[0.88, 0.85, 0.15]
	9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
	10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]

Result Grid Form Editor Field Types

Step 9 — Similarity Search: Cosine Similarity

```

SELECT
    book_id, title, author, genre,
    ROUND(
        (
            CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) * 0.90 +
            CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) * 0.80 +
            CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) * 0.20
        ) /
        (
            Sqrt(
                POW(CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)),2) +
                POW(CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)),2) +
                POW(CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)),2)
            )
            * Sqrt(POW(0.90,2) + POW(0.80,2) + POW(0.20,2))
        ), 4
    ) AS cosine_similarity
FROM Books
WHERE embedding IS NOT NULL
ORDER BY cosine_similarity DESC
LIMIT 3;

```

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23
24
ion25
26
27
28
29
30
31

```
)
    * SQRT(POW(0.90,2) + POW(0.80,2) + POW(0.20,2))
    ), 4
) AS cosine_similarity
FROM Books
WHERE embedding IS NOT NULL
ORDER BY cosine_similarity DESC
LIMIT 3;
```

100% 25:27

Result Grid

Filter Rows: Search

Export:

Fetch rows:

	book_id	title	author	genre	cosine_similar...	
	1	The Martian	Andy Weir	Science Fiction	1	
	2	Dune	Frank Herbert	Science Fiction	0.9986	
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983	